Tour de maths : Junior competition

Redhill School : 2014

Question 1
A (-7 ;4) and B (3 ; -9) are two points in a set of axes, find the length of AB.

Question 2
The product of two numbers is 504 and each of the numbers is divisible by 6. Neither of the two numbers is 6. What is the larger of the two numbers?

Question 3
Given : the following 6 numbers

11; 31; 19; 3; 10; 6

Three numbers are selected from the above set and added together. The remaining three numbers are also added together. These possible two sums are then multiplied to get a product. What is the maximum product?

Question 4
Find the value of

2002-2001+2000-1999+......................+2-1

Question 5
A cyclist notices that his average speed when he has covered exactly half the total distance of his race is 30 km/h. What should his average seed be over the second half of the race if he wants to finish the race with an average speed of 40 km/h?

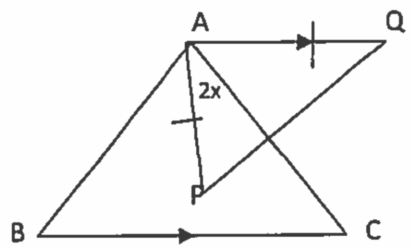
Question 6
What is the remainder when 7^{2014} is divided by 10?

Question 7

What is the smallest integer x, for which $\frac{15}{x-1}$ is an integer?

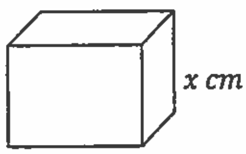
Question 8 Find in terms of x a three digit number that has x as its unit digit, $(x-1)$ as the tens digit and $\frac{1}{2}x$ as its hundreds digit.

Question 9



In the figure above, ΔABC is equilateral and ΔAPQ is isosceles with AQ=AP and AQ//BC.
If $P\widehat{A}C$ = 2x, find $\hat{Q}$ in terms of x.

Question 10



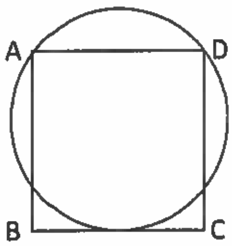
If a cube has a side $x\ cm$. Find the value of this side if the volume of the cube is the same as the total surface area.

Question 11
1000 dots are evenly spaced on the circumference of a circle. They are numbered from 1 to 1000 with dot 1000 opposite dot 500. Which dot is opposite 657?

Question 12
Each male Honey-bee has a single female parent whilst each female honey-bee has both a male and a female parent. In the 10th generation back,only, how many ancestors does a male honey-bee have?

Question 13
There are five sticks measuring 1 cm, 2 cm, 3 cm, 4 cm and 5 cm. How many different triangles can be formed using three sticks at a time.

Question 14



A circle passes through vertices A and D and touches side BC of a square. If the square has side length 2, what is the radius of the circle?

Question 15

If $|x| = \begin{matrix} x & if\ x \geq 0 \\ -x & if\ x < 0 \end{matrix}$ and $a < b < c$

Find
$|a - b| + |b - c| - |c - a|$

Question 16

A dice is thrown twice. The first throw determines the tens digit and the second throw determines the unit digit of a two digit number. What is the probability that this two digit number is a perfect square?

Question 17

John can build a wall in 16 hours if he works alone. Jane can build the same wall in 12 hours if she works alone. If they work together they can build the wall in 8 hours, but because they sometimes get in each others way, they build 16 bricks less per hour than they would if they did not get in each other's way. How many bricks are there in the wall?

Question 18

Five straight lines are drawn. What is the maximum number of points of intersection?